

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1711

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

I Kishore S.P (192511081) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications*.

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

Task 1: Article Summary

(a) Citation and problem

The paper develops a **reinforcement-learning-based personalised diabetes treatment recommendation system** using South Korean EHR data. The main problem is that traditional MDP-based treatment models can represent only a limited number of patient states and actions because of the high dimensionality of medical information.

(b) Objective

The authors aim to improve personalised diabetes treatment recommendations using patient-specific EHR information. Traditional approaches often use a limited state representation, which may ignore important clinical details. The research therefore addresses the gap between simple RL treatment models and the complex, high-dimensional information available in real-world electronic health records. The authors propose an **expanded personalised model based on contextual bandits**, allowing more patient information to be incorporated into the decision process. The model uses medical history, medications, laboratory tests and other patient information to recommend appropriate diabetes treatment. It is designed to assist physicians rather than simply provide a general "one-size-fits-all" treatment. The proposed approach is compared with existing reinforcement-learning algorithms, and the authors report improved performance. The study demonstrates how sequential decision-making and patient-specific information can be combined to support precision medicine and diabetes treatment decisions.

Task 2: Methodology Analysis

(a) Technique and dataset

The authors use:

- **Reinforcement Learning**
- **Contextual Bandits**
- **Electronic Health Records**
- Patient medical history
- Medication information
- Laboratory-test information

The data came from **South Korean national health-insurance/EHR records**. The model expands the patient state representation so that more clinically relevant information can be considered.

(b) CO4 mapping

CO4 Topic	Connection
Reinforcement Learning	Agent learns actions that maximise cumulative reward.
MDP	Patient state, action and outcome form the decision process.
Inductive Learning	Historical patient data are used to learn useful treatment patterns.

Strength: Personalises recommendations using more patient information than a simple MDP.

Limitation: Results based on retrospective EHR data may not guarantee that the recommended treatment will produce the same outcome in real patients.

Task 3: Results & Critical Evaluation

(a) Results

The authors report that the proposed contextual-bandit approach **outperformed existing reinforcement-learning models** and improved diabetes treatment recommendation performance.

Method	Result
Existing RL methods	Lower performance
Proposed contextual bandit	Best performance

The article's main conclusion is that the expanded model can improve personalised treatment recommendations.

(b) Critical evaluation

The results are promising, but:

- EHR data may contain missing or biased information.
- Retrospective evaluation is weaker than prospective clinical trials.
- Treatment recommendations may be affected by physicians' historical choices.
- External validation using another country's dataset would strengthen the study.

(c) Real-world applicability

The technique could support:

- Diabetes medication selection
- Personalised treatment planning
- Clinical decision-support systems
- Hospital/EHR applications

Major challenges include **patient privacy, bias, explainability, safety, data quality and clinician acceptance**.

Task 4: Reflection

(a) Three key insights

1. **RL can solve sequential medical decisions** rather than only predicting disease.
→ CO4: Reinforcement Learning.
2. **Patient state representation is extremely important.** More relevant features can improve personalised recommendations.
→ CO4: MDP/State representation.
3. **Historical healthcare data can be used to learn treatment policies**, but safety and validation are essential.
→ CO4: Inductive Learning + RL.

(b) Proposed extension

I would test the model on a **large international diabetes EHR dataset** and compare it with Deep Q-Learning.

This would test whether the learned policy generalises across different populations and healthcare systems, improving reliability and reducing dataset-specific bias.

Conclusion

The article is highly relevant to CO4 because it demonstrates how **Reinforcement Learning and contextual bandits can transform historical EHR data into personalised diabetes treatment recommendations**. Its major contribution is moving from simple patient-state models toward more detailed, personalised clinical decision-making.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand